

1 Optimization Strategy

The task was treated as a constrained black-box optimization problem, since each objective evaluation requires a full FreeCAD FEM simulation. The design vector is

$$\mathbf{x} = (x_1, y_1, x_2, y_2, x_3, y_3, \alpha),$$

where (x_1, y_1) and (x_2, y_2) define the two circular cutouts, while (x_3, y_3) and α define the position and rotation of the elongated cutout. The optimization objective is to maximize the absorbed deformation energy

$$\max_{\mathbf{x}} E_{\text{abs}}(\mathbf{x})$$

subject to the geometric feasibility constraints and the stress constraint

$$\sigma_{\text{max}}(\mathbf{x}) \leq 275 \text{ MPa.}$$

The parameter bounds in `config.json` were chosen to cover a large practically useful design space while already excluding some obviously infeasible edge cases. The final bounds were

$$\begin{aligned} x_1 &\in [11, 49], & y_1 &\in [11, 19], \\ x_2 &\in [8, 52], & y_2 &\in [8, 52], \\ x_3 &\in [9.5, 50.5], & y_3 &\in [9.5, 50.5], \\ \alpha &\in [0, 180]. \end{aligned}$$

Since many samples in this space still violate the geometric constraints, samples were filtered before running the FEM simulation. The implemented checks enforce the minimum edge and inter-object distance.

Initial data was generated using Latin Hypercube Sampling (LHS). Each valid sample was evaluated with the FreeCAD FEM model, and the design variables, absorbed energy, maximum stress and timestamp were stored in a CSV file. Saving all valid simulations was important because every FEM run is expensive and can later be reused as training data for the surrogate models. Two separate Gaussian Process surrogate models were trained: one for the absorbed energy and one for the maximum stress. For every candidate design $\mathbf{x}$, the energy model predicts a mean value $\mu_E(\mathbf{x})$ and uncertainty $\sigma_E(\mathbf{x})$, while the stress model predicts $\mu_\sigma(\mathbf{x})$ and $\sigma_\sigma(\mathbf{x})$. Separate length scales were used for all seven design variables, since coordinates and rotation angle influence the geometry in different ways.

The optimization then followed a Bayesian optimization strategy. A large set of new candidate designs was generated with LHS and filtered for geometric validity. Instead of evaluating all candidates with FEM, the GP predictions were combined a new score, which combines an upper confidence bound acquisition function with a soft stress penalty

$$a(\mathbf{x}) = (\mu_E(\mathbf{x}) + \beta\sigma_E(\mathbf{x})) \cdot \exp\left(-\frac{1}{2} \max\left(\frac{\mu_\sigma(\mathbf{x}) - 275}{\sigma_\sigma(\mathbf{x})}, 0\right)^2\right),$$

with $\beta = 1$.

The newly evaluated candidates were appended to the CSV dataset and included in the next GP training run. While this is not a fully automated online Bayesian optimization loop, it follows the same sequential principle. Since the data storage and retraining steps are already implemented, the workflow could be extended to a fully automatic loop by adding an outer iteration.

This approach was preferred over direct gradient-based optimization of the GP surrogate. Although the surrogate is cheap to evaluate, direct optimization can exploit local surrogate artefacts

and produce candidates that are unreliable under the true FEM model. Sampling many valid candidates and ranking them with the surrogate-based acquisition function provided a simple and robust procedure while keeping the number of FEM evaluations limited.

The convergence behavior is shown in Fig. 1, which plots the best feasible sample found in each run over time. Fig. 3 shows all FEM-evaluated samples in the energy–stress plane and indicates which samples satisfy the stress constraint. Fig. 2 shows the transition from the initial data generation phase to the subsequent Bayesian optimization runs. The later samples increasingly satisfy the stress constraint while also showing an upward trend in absorbed energy. The final best feasible design achieved an absorbed energy of **13228.69** at a maximum stress of **235.56** MPa after **1184** FEM evaluations.

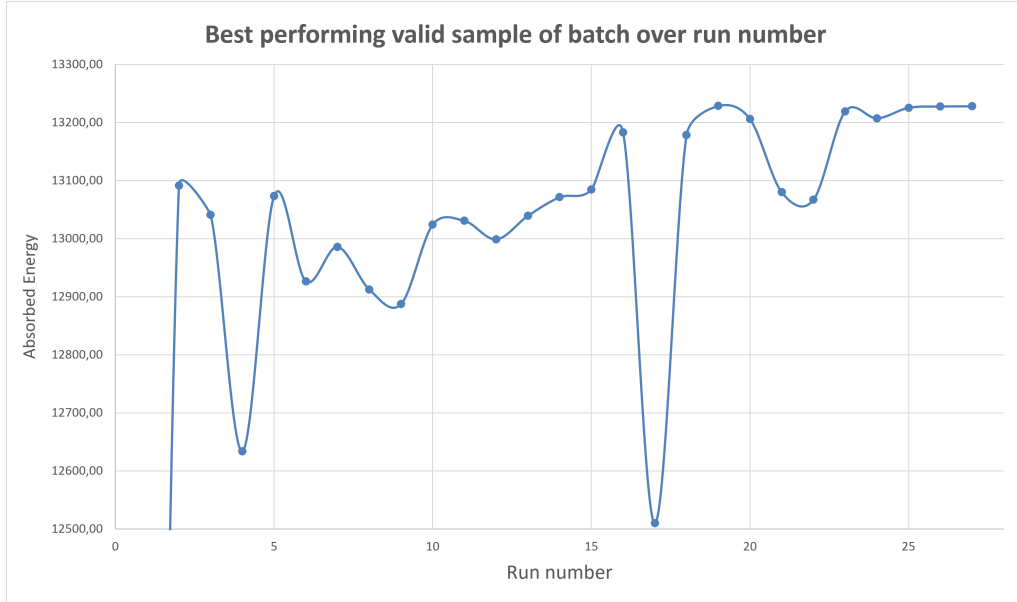


Figure 1

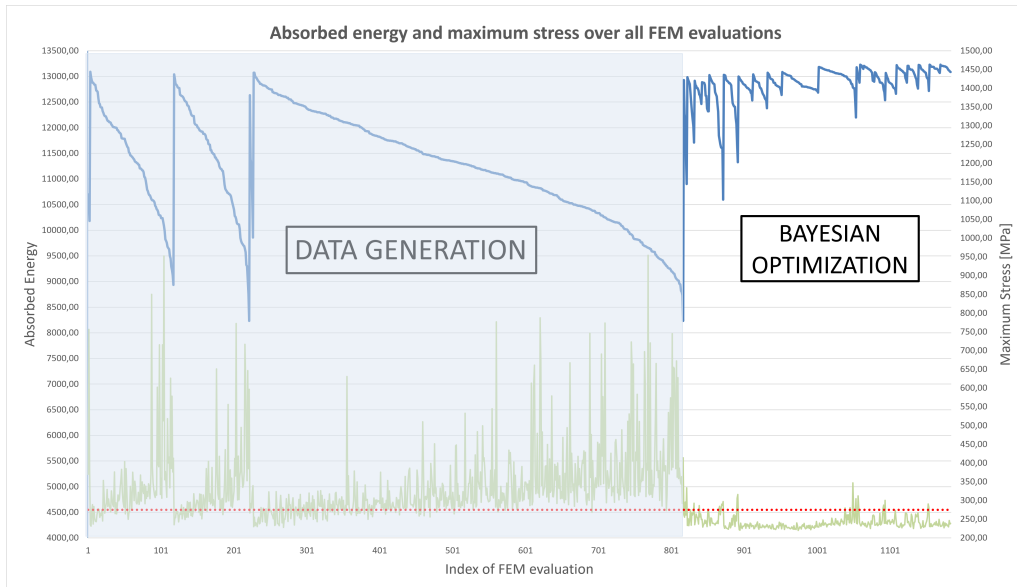


Figure 2: Blue: Absorbed Energy, Green: Max. Stress, Red: $\sigma = 275$ MPa constraint

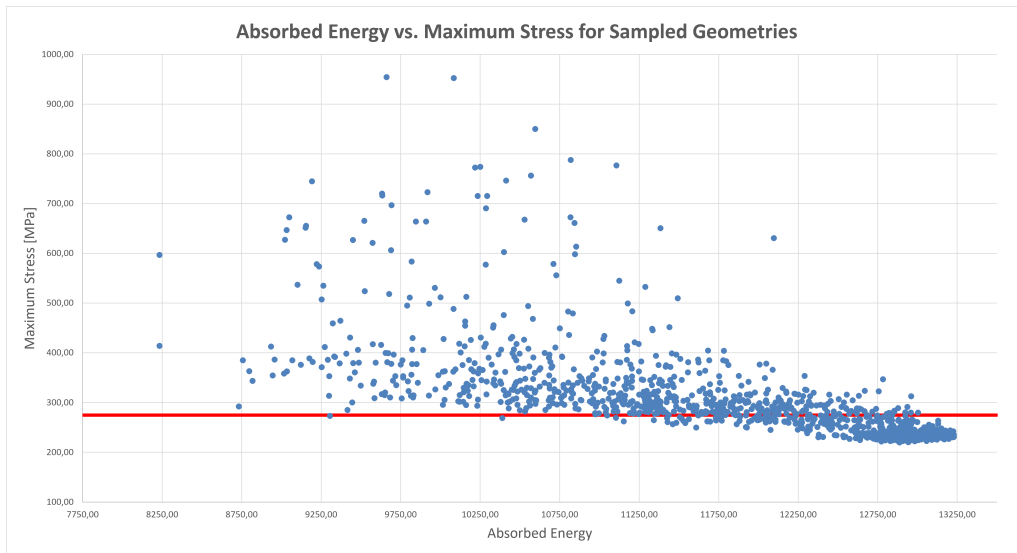


Figure 3: Red line indicating $\sigma = 275$ MPa constraint

Finally, this was the optimal design found by the optimization [4](#)

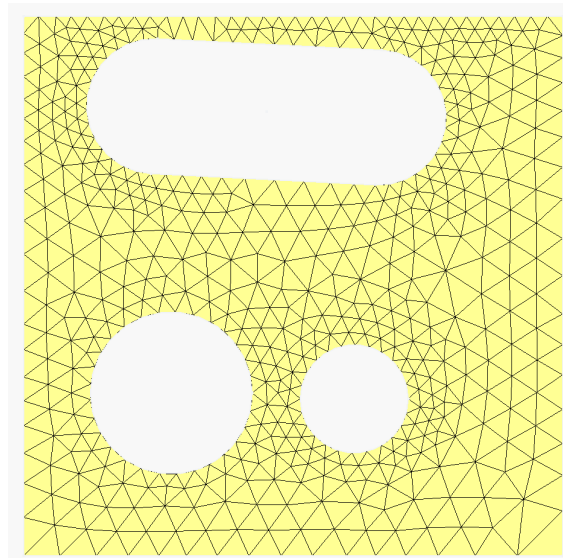


Figure 4: Best performing sample